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NIRGB-GS: Near-Infrared Assisted Low-Light Scene Reconstruction and Enhancement via Gaussian Splatting

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ABSTRACT

3D Gaussian Splatting (3DGS) has emerged as a powerful framework for scene reconstruction and novel view synthesis for normal-light and illumination-consistent scenes. However, the original 3DGS struggles to reconstruct scenes from low-light RGB images, which are characterized by low pixel intensity and significant noise degradation. Near infrared (NIR) images captured under invisible near-infrared light can maintain high signal-to-noise ratio and brightness in extremely dark environments but lack color information and suffer from interimage illumination inconsistency. In this article, we propose NIR-assisted low-light RGB Gaussian Splatting (NIRGB-GS), a multimodal 3DGS-based method for 3D reconstruction and normal-light novel view synthesis of low-light scenes. We adopt an NIR-dominated Gaussian density control strategy to mitigate the influence of noise in RGB images and address the lighting inconsistency in NIR images with appearance encoding. Inverse histogram equalization is proposed for color and brightness recovery, even in extreme low-light conditions. Experimental results across 7 real-world low-light scenes demonstrate that NIRGB-GS is able to recover geometry, color, and texture of low-light scenes and outperforms existing algorithms in both subjective perception and objective evaluation.

1 | Introduction

Scene reconstruction and novel view synthesis are essential tasks in 3D vision with promising applications in autonomous driving [1–3], virtual reality [4–6], and robotics [7–10]. Recently, Neural Radiance Fields (NeRF) [11] and 3D Gaussian Splatting (3DGS) [12] have emerged as transformative frameworks, enabling highly detailed scene reconstruction and photorealistic novel view rendering requiring only multiview images and camera poses. NeRF implicitly represents a scene as a continuous volumetric scene function that models the color and density of sampled points within the scene. 3DGS employs explicit and discrete Gaussian primitives to represent the scene and leverages differentiable rasterization for real-time rendering, significantly improving training and rendering efficiency compared to NeRF.

However, 3DGS struggles in low-light environments [13–15]. Images captured in dark scenes typically exhibit significant noise and low average brightness [16], creating a substantial challenge for 3D reconstruction with 3DGS. (1) Training 3DGS requires accurate camera poses, typically obtained using Structure from Motion (SfM) [17] methods like COLMAP. However, the low brightness and high noise levels in low-light images frequently cause SfM to fail [18]. (2) Directly applying 3DGS to low-light RGB imagery naively reconstructs the low illumination and noise of the inputs, producing synthesized novel views that are dark and contaminated with visual artifacts such as floaters.

An intuitive approach is to apply 2D low-light enhancement (LLE) to dark RGB images before feeding them into the 3DGS pipeline. However, this per-image enhancement often disrupts illumination consistency across the dataset, leading to color

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distortion and ghosting artifacts in the final reconstruction and synthesized views [19]. Another straightforward strategy is to use the camera built-in flash, which brightens the scene during capture. However, this strategy inherently creates illumination inconsistencies across frames due to the changing light source. Furthermore, camera flashes can be disruptive to people present in the environment.

3DGS-based approaches specifically designed for low-light conditions have been proposed. DarkGS [20] models the visible camera-light system and utilizes the model for scene reconstruction from partially illuminated images. However, this method relies on raw images and requires manual calibration and white balance. Luminance-GS [21] adds color adjustment parameters within 3DGS framework for scenes reconstruction under challenging lighting conditions. The adjustment parameters are guided by per-view color matrix mapping and view adaptive curve adjustments for lighting adaptation and cross-view consistency. LLGS [22] introduces a novel absorptance attribute for each 3D Gaussian primitive to model light behavior in low-light conditions and omits the absorptance for normal-light novel view synthesis. Despite these advancements, existing approaches still face challenges in reconstructing low-light scenes from monomodal RGB images. They exhibit significant difficulty in recovering texture and boundary details in extremely dark regions, such as the corner area in Figure 1a. Even the state-of-the-art (SOTA) methods specifically tailored for low-light 3DGS, such as Luminance-GS and LLGS, struggles to fully recover fine details and natural colors, as evidenced by the obscure details and distorted colors in Figure 1c,d.

Multispectral fusion for 3D reconstruction has emerged as a promising solution. This paradigm aims to enhance the reconstruction quality of a target modality by integrating complementary data from an auxiliary modality. Current multispectral NeRF and 3DGS research primarily focuses on fusing visible and thermal infrared spectra images. Thermal-NeRF [23] fuses thermal infrared images and low-light RGB images for 3D scene reconstruction. It leverages the illumination invariance of thermal images to recover the general geometric information of a scene under extremely dark conditions. 3DGS-based ThermalGaussian [24] enhances infrared reconstruction quality using normal-light RGB images. Although thermal infrared imaging is robust in low-light environments, its application in enhancing low-light RGB reconstruction still faces practical limitations: thermal sensors are generally expensive for broad deployment [25] and suffer from low resolution [26, 27].

Conversely, near-infrared (NIR) imaging captured with active near-infrared light offers a compelling and low-cost dark-environment sensing modality. Critically, NIR light (>700 nm) is invisible to human eyes, making this illumination completely non-disruptive to people [28]. NIR images typically exhibit high brightness and significantly higher signal-to-noise ratio compared to visible-light RGB images in low-light scenarios [29–31] as shown in Figure 1a and Figure 1b. However, NIR data introduces its own set of challenges: it inherently lacks color information crucial for realistic scene rendering, and also introduces interimage illumination inconsistency.

To overcome these challenges and accomplish high-fidelity 3D reconstruction in low-light scene, we propose NIR-assisted low-light RGB Gaussian Splatting (NIRGB-GS), a novel framework that fuses NIR and low-light RGB images within the 3DGS paradigm. Our core insight is to leverage the complementary strengths of both modalities: utilizing the high-SNR NIR imagery to robustly guide scene geometry estimation and mitigate the impact of RGB noise, while recovering the color information from the low-light RGB captures.

Specifically, we use appearance encoding to model the illumination variations inherent in NIR sequences. We adopt a strategy where NIR information dominates the Gaussian density optimization. We propose inverse histogram equalization (IHE) to effectively restore natural color from the noisy low-light RGB inputs, enabling the synthesis of normal-light novel RGB views. Our framework effectively reconstructs the scenes even for areas completely dark in the RGB images by relying on the corresponding NIR modality, as shown in Figure 1e. To the best of our knowledge, NIRGB-GS is the first work to successfully integrate NIR-assisted multimodal fusion into the 3DGS framework to tackle the challenge of extremely low-light scene reconstruction and enhancement. Comprehensive experiments demonstrate that NIRGB-GS significantly outperforms existing state-of-the-art methods for low-light scene reconstruction, achieving superior results in both subjective visual quality and objective metrics.

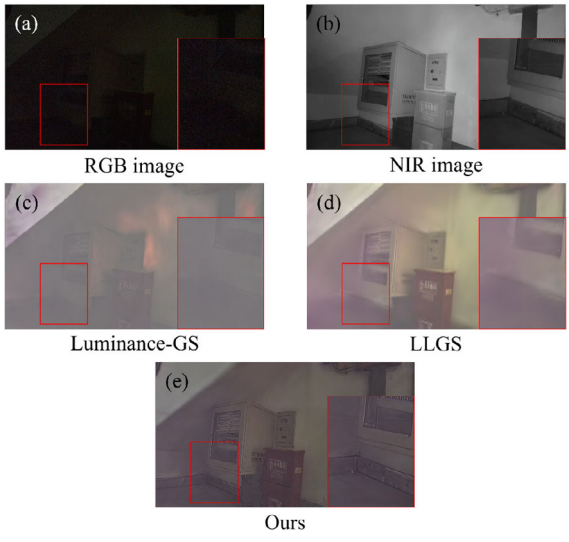


FIGURE 1 | RGB/NIR ground truth images and comparison of different reconstruction methods. (a) Low-light RGB image (noisy and dark). (b) NIR image at the same camera pose (high brightness and clear structure). (c) Result of Luminance-GS [21]. (d) Result of LLGS [22]. (e) Result of our proposed method.

2 | Method

2.1 | 3DGS Preliminaries

3DGS represents the scene as a collection of 3D Gaussian primitives. Each Gaussian primitive possesses 4 types of parameters: the center $\boldsymbol{\mu}_i \in \mathbb{R}^3$, covariance matrix $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$, opacity $\alpha_i \in [0, 1]$, and view-dependent color represented by spherical harmonics (SH) coefficients. The geometric representation of each Gaussian is defined by the following equation:

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$$G_i(\boldsymbol{x}) = \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^{\mathrm{T}}\boldsymbol{\Sigma}_i^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right) \tag{1}$$

In practice, the covariance matrix $\boldsymbol{\Sigma}_i$ is decomposed into scaling matrix $\mathbf{S}_i \in \mathbb{R}^{3\times3}$ and rotation matrix $\mathbf{R}_i \in \mathrm{SO}(3)$ through:

$$\boldsymbol{\Sigma}_i = \mathbf{R}_i\mathbf{S}_i\mathbf{S}_i^{\mathrm{T}}\mathbf{R}_i^{\mathrm{T}} \tag{2}$$

In the vanilla 3DGS, the color of a Gaussian primitive $\boldsymbol{c}_i \in \mathbb{R}^3$ along the observation direction $\boldsymbol{v}_i \in \mathbb{R}^3$ can be expressed through 3rd-order SH coefficients $\boldsymbol{k}_l^m \in \mathbb{R}^{48}$ and SH basis functions Y_l^m, the formula is as follows:

$$\boldsymbol{c}_i = \sum_{l=0}^{L}\sum_{m=-l}^{l}\boldsymbol{k}_l^m Y_l^m(\boldsymbol{v}_i) \tag{3}$$

When rendering, 3DGS projects 3D Gaussian primitives to a target 2D plane and blends multiple overlapping depth-ordered projections to form the color $\boldsymbol{C} \in \mathbb{R}^3$ of associated pixels as:

$$\boldsymbol{C} = \sum_{i\in N}\boldsymbol{c}_i o_i \prod_{j=1}^{i-1}(1-o_j), \quad \text{where} \quad o_i = \alpha_i G_i^{2\mathrm{D}} \tag{4}$$

where N is the set of depth-ordered Gaussians overlapping the pixel, and $G_i^{2\mathrm{D}}$ is the 2D projection of the 3D Gaussian onto the image plane.

2.2 | NIRGB-GS Framework

The workflow of NIRGB-GS is illustrated in Figure 2. First, the NIR images are processed through SfM method like COLMAP to obtain camera poses and sparse point clouds, detailed in Section 2.3. Sparse points are employed to initialize the position (center $\boldsymbol{\mu}$) and shape information (covariance matrix $\boldsymbol{\Sigma}$) of the 3D Gaussian primitives. These two parameters are shared across NIR and RGB modalities, ensuring a consistent geometric representation in the 3D space. To capture modality-specific visual characteristics, the two modalities are assigned separate color properties $\boldsymbol{c}$ and opacity features α. Specifically, color-related parameters are obtained through separate appearance models, detailed in Section 2.4. Then, the two shared and two separate parameters are used to render the normal-light and NIR images respectively. These rendered images are then evaluated using a loss function with three components: the pixel-wise loss between IHE (detailed in Section 2.5) transformed normal-light render images (low-light render images) and RGB ground truth (GT), the structural similarity index measure (SSIM) loss between normal-light render images and NIR GT, and the SSIM loss between the NIR render images and NIR GT. Noticeably, density control of the 3D Gaussians is determined by NIR opacity α_2, detailed in Section 2.6.

2.3 | NIR COLMAP

The noisy and low-intensity RGB images captured under low-light conditions often lead to COLMAP failure. Since COLMAP internally converts all images to grayscale prior to keypoint detection and feature matching [32], the absence of color information in NIR images does not present a disadvantage. Crucially, NIR images maintain high brightness and signal-to-noise ratio (SNR) under low-light conditions, enabling successful camera pose estimation with COLMAP. A representative example is provided in Figure S1 in the Supporting Information.

Within our dataset, each RGB image is paired with a corresponding NIR image captured from the same camera pose. Consequently, the accurate NIR camera poses set can be directly applied to their paired RGB counterparts. This provides NIRGB-GS pipeline with reliable camera poses and initial point cloud geometry.

Compared to other methods for low-light reconstruction, our approach demonstrates superior robustness and operational convenience for practical deployment. Aleth-NeRF [33] collects the LOM dataset containing paired low-light and normal-light images. The low-light images are captured in normal-light scenes by adjusting exposure time and ISO of the camera. The camera poses of the images are attained by adopting COLMAP on the normal-light images. However, this pipeline is impractical in genuine dark environments. LLNeRF [34] processes images captured in dark environments with 2D image enhancement techniques and then performs image pose estimation. The enhancement process generally preserves or even amplifies noise while introducing inconsistencies across the dataset,

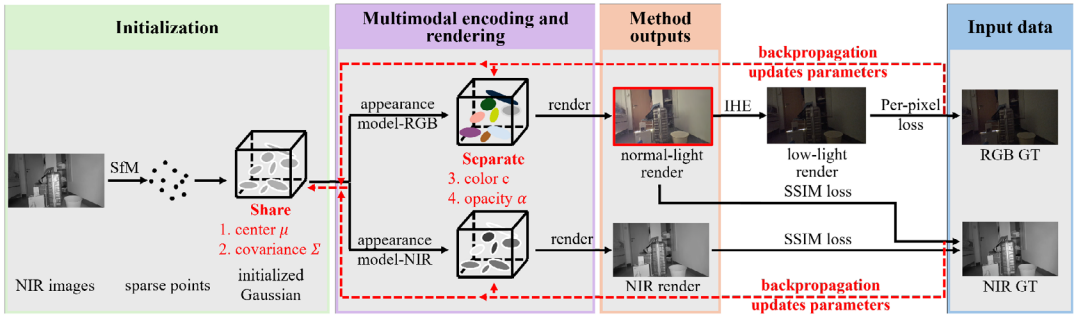


FIGURE 2 | Overview of our NIRGB-GS pipeline. (1) The NIRGB-GS pipeline begins with processing NIR images via SfM to obtain sparse points cloud and camera poses. Sparse points initialize the 3D Gaussian primitives with two shared parameters. (2) The NIR and RGB modalities each have their own color and opacity parameters. The separate color features are obtained through separate appearance models. (3) The shared and separated parameters are used to rasterize normal-light and NIR render outputs. (4) The rendered images are then evaluated using a total loss function with three loss components. The loss gradients are backpropagated to update and optimize the 3D Gaussians' parameters (including centers $\boldsymbol{\mu}$, covariance matrices $\boldsymbol{\Sigma}$, RGB opacity $\boldsymbol{\alpha}_1$, and NIR opacity $\boldsymbol{\alpha}_2$), and 2 appearance models.

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therefore compromising pose estimation accuracy and incurring additional computational overhead. In contrast, our strategy avoids these limitations by directly utilizing NIR images, achieving a high COLMAP success rate without requiring simultaneous normal-light capture or preprocessing steps.

2.4 | Appearance Model

The NIR lights are positioned on both sides of the camera, as illustrated in Figure 3a. During multiview capture, camera movement inevitably leads to variations in NIR illumination across different frames. For example, the door appears dark in one NIR image Figure 3b, but becomes bright in another frame when the camera orientation directs the lights toward it, as shown in Figure 3c.

Furthermore, when capturing RGB images in low-light environments, the camera white balance becomes highly sensitive to minor illumination variations, often causing significant color shifts between different frames (see Figure S2). The illumination inconsistencies in NIR datasets and white balance fluctuations in low-light RGB datasets introduce interimages variations that pose significant challenges for 3D scene reconstruction.

Inspired by the appearance modeling in Splatfacto-W [35], we address the above issues by introducing two appearance model modules to describe the per-image inconsistencies for NIR and RGB modalities separately. Each appearance module assigns a per-image 48-dimensional appearance embedding and a per-Gaussian 72-dimensional appearance feature vector. These two inputs are processed through an MLP to calculate SH coefficients, which explicitly model frame-specific variations in lighting and color response (see Figure S3). Experimental validation confirms that this strategy effectively eliminates reconstruction floaters and artifacts, as detailed in the ablation study.

2.5 | Inverse Histogram Equalization

Since our objective is to reconstruct scenes under normal-light conditions while the RGB training dataset comprises only low-light images, directly computing a loss function between predictions and low-light RGB GT is meaningless. We transform the normal-light render image via IHE. This operation remaps pixel intensities from the standard 0–255 range to the specific intensity distribution of our low-light dataset, as formalized in Equation (5):

$$\boldsymbol{l}_{\text{dark}} = \frac{k_{\max} - k_{\min}}{255 - 0} * \boldsymbol{l}_{\text{normal}} \tag{5}$$

where $\boldsymbol{l}_{\text{dark}}$ denotes the low-light render image, $\boldsymbol{l}_{\text{normal}}$ represents the normal-light render image, $k_{\min}$ is the lower bound of the target intensity range (fixed at 0), and $k_{\max}$ is the upper bound corresponding to the 99th percentile of the cumulative pixel intensity distribution aggregated across the entire low-light training dataset (see discussion on the percentile parameter in Section S6). We then compute a pixel-wise loss L_{RGB} between IHE transformed prediction and the low-light ground-truth images.

2.6 | Density Control

In our framework, the Gaussian density control strategy adopts the same core mechanism as 3DGS [12], encompassing Gaussian pruning and densification to optimize the number and distribution of Gaussian primitives. The entire density adjustment process is exclusively governed by the NIR opacity α_2 and positional gradients. Specifically, Gaussians with NIR opacity α_2 below a threshold are pruned as they contribute minimally to the scene structure; in contrast, those with view-space positional gradients exceeding a threshold are densified to enhance the representation of local details. The densification operation includes splitting and cloning: large Gaussians with high variance are split into smaller ones, while small Gaussians with low variance are cloned.

This density control strategy leverages the high SNR and stable structural information in NIR images. With NIR opacity α_2, rather than low-light RGB opacity α_1, dominating Gaussian pruning, we achieve stable geometries and significant artifact suppression in the reconstructed scene.

2.7 | Loss

The loss function of NIRGB-GS is represented as Equation (6):

$$L = \lambda_1 L_{\text{RGB}} + \lambda_2 L_{\text{SSIM-RGB}} + \lambda_3 L_{\text{SSIM-NIR}} \tag{6}$$

L_{RGB} denotes the pixel-wise loss between the IHE transformed normal-light render images (low-light render images) and the low-light RGB ground truth. This loss ensures that the low-light render images closely match the low-light RGB ground truth, thereby guiding the model to learn accurate color information from the available RGB images.



FIGURE 3 | NIR dataset illumination inconsistency demonstration. (a) Our image acquisition device (Ezviz C6WI) equipped with fixed infrared illuminators on both sides (highlighted by red boxes). (b) A NIR frame of our bedroom scene. In this frame, the door appears dark, as the camera orientation does not direct NIR lights toward it. (c) The door becomes bright in another NIR frame of the same scene, due to camera motion aligning NIR lights with the door. This shows variable illumination across NIR frames caused by changing camera orientation relative to the built-in NIR light sources.

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$L_{\text{SSIM}-\text{RGB}}$ quantifies the SSIM loss between the normal-light render images and the NIR ground truth. This cross-modal loss enforces structural similarity between the RGB rendering and NIR ground truth, harnessing the high-SNR NIR information to bolster structural fidelity within synthesized RGB outputs, particularly in extremely dark regions where RGB ground truth images hardly provide any information.

$L_{\text{SSIM}-\text{NIR}}$ denotes the SSIM loss between the NIR render images and the NIR ground truth. This loss ensures high-fidelity reconstruction of the NIR modality by maintaining structural consistency with NIR inputs. This is crucial because the positions and shapes of Gaussian primitives are shared between both modalities. Furthermore, our NIR opacity-dominated Gaussian pruning strategy implicitly regularizes the shared 3D geometry, which in turn improves the overall quality and convergence speed of the RGB reconstruction.

The loss weights λ_1, λ_2, and λ_3 balance the loss components, with values empirically set to 2, 0.2, and 1 respectively. As detailed in Section 3.3.1, we systematically analyze how variations in λ_1 affect RGB reconstruction quality. Furthermore, in Section 3.3.2, we discuss the impact of removing the critical cross-modal loss $L_{\text{SSIM}-\text{RGB}}$.

3 | Experiments

3.1 | Datasets and Comparisons

3.1.1 | Implementation Details

We minimize the total loss to optimize the 3D Gaussians parameters along with the 2 MLP weights, appearance embeddings, and Gaussian appearance features. We train 10 000 iterations on a single Nvidia RTX A6000. The appearance embeddings and Gaussian appearance features are respectively 48 and 72 dimensional. The appearance model is a three-layer MLP of width 256. All parameters are optimized using the Adam optimizer, with key learning rates as follows: the learning rate for appearance embeddings starts at 1e-3 and decays to 3e-4, and the learning rate for Gaussian appearance features starts at 2e-2 and decays to 1e-3. The learning rates for all other parameters are consistent with our baseline method Splatfacto [36].

Pruning and densification are performed every 100 iterations after a warm-up stage. We also reset the opacity every 2500 iterations to stabilize densification statistics. At each refinement, we prune Gaussians whose NIR opacity α_2 falls below 0.1, and remove oversized primitives whose maximum scale exceeds 0.5. We additionally cull Gaussians whose projected size is larger than 0.15 of the image resolution. For densification, Gaussians are selected when the view-space positional gradient norm exceeds 8e-4; selected Gaussians are split if their projected size exceeds 0.05 of the screen space, otherwise they are duplicated. These settings are default settings of Splatfacto [36] and we keep them unchanged in this article.

3.1.2 | Datasets

Our dataset was captured using a consumer-grade surveillance camera (Ezviz C6WI) with built-in NIR illuminators. This device enables sequential acquisition of both RGB and NIR images through manual switching while maintaining a fixed position, ensuring strict pose correspondence for each RGB-NIR image pair. We collected data in 7 challenging real-world low-light environments: bedroom, living room, tractor, study, entrance hallway, HVAC room, and gate. We reserved 10% of the data as the test set to evaluate model generalization, using the remaining 90% for training.

3.1.3 | Comparison Methods

To evaluate the effectiveness of our approach in restoring high-quality normal-light 3D scenes from dark environments, we compare it against several baseline methods. In particular, we include two 3DGS-based low-light reconstruction methods: Luminance-GS [21] and LLGS [22]. We further evaluate a pipeline method termed DVN + Splatfacto. In this setting, the 2D RGB-NIR fusion low-light enhancement method DVN [29] is first applied to dark RGB images, which leverages paired NIR images to enhance low-light RGB inputs. The enhanced images are then fed into Splatfacto [36], the nerfstudio implementation of 3DGS. To validate the effectiveness of integrating NIR information, we introduce a fourth baseline, Splatfacto_IHE, adapted from the original Splatfacto designed for normal-light RGB images. We extend it to multispectral reconstruction by sharing Gaussian parameters between the NIR and RGB modalities while only keeping the color-related parameters separate. To further adapt this method to dark environments, we integrate our proposed IHE module into the pipeline to restore proper illumination.

3.1.4 | Metrics

To assess the quality of low-light scene reconstruction and enhancement, we first compare the novel-view synthesis images against corresponding reserved low-light ground truth images (withheld from the training dataset). Since the low-light condition reduces edge features and detail information in images, the clarity of scene restoration can be measured by the edge strength and detail contrast in novel view synthesis images. An effective evaluation method is the visible edge gradient approach [37]. This technique quantifies the enhancement effect using 2 metrics: newly visible edges ratio (e) and average gradient ratio ($\bar{r}$), as formalized in Equation (7):

$$e = \frac{n_r - n_0}{n_0}, \quad \bar{r} = \frac{\bar{g}_r}{\bar{g}_0} \tag{7}$$

where n_0 and n_r respectively represent the number of visible edges in the dark RGB images and the novel-view synthesis images, $\bar{g}_0$ represents the average gradient of the dark RGB images, $\bar{g}_r$ represents the average gradient of the novel-view synthesis images.

Additionally, we evaluate the practicality of the proposed method through ORB feature matching. Specifically, we extract 1,000 ORB features from enhanced novel-view synthesis images of two perspectives, then perform feature matching between each ORB features. Following the evaluation criteria of ORB-SLAM3 [38], we quantify the number of high quality matches by applying two strict conditions: (1) the Hamming distance must be less than or equal to 50; (2) the best match distance must be less than 0.6 times the second-best match distance. By quantifying the number of high quality matches under these criteria, we investigate the effectiveness of NIRGB-GS in

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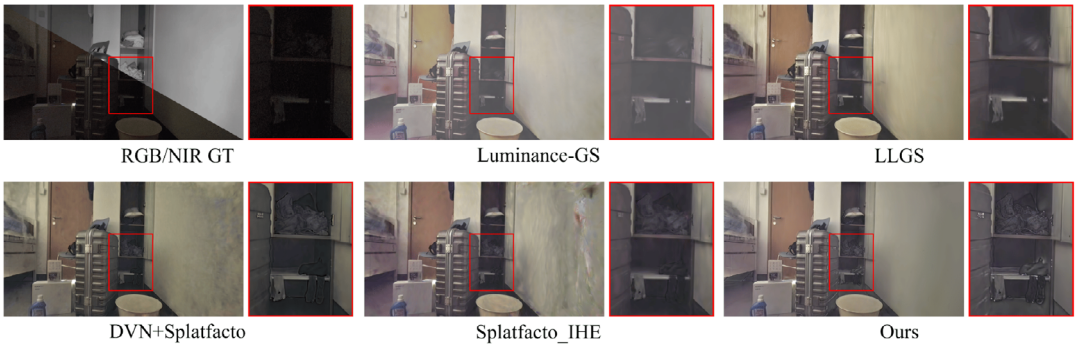


FIGURE 4 | Results of the bedroom scene. The magnified RGB GT images show that the wardrobe area is too dark without sufficient valid pixel information. Consequently, RGB-based low-light enhancement reconstruction methods (Luminance-GS and LLGS) struggle to reconstruct texture details here. The 3 NIR-RGB-fusion methods below can reconstruct objects here, with our method achieving the best reconstruction results. For instance, our method can clearly reconstruct the patterns on the shirt.

supporting downstream perception tasks (e.g., SLAM, obstacle detection), which is critical for robotic applications in low-light environments.

3.2 | Experiment Results

3.2.1 | Visual Comparison

As illustrated in Figures 4–6, our method demonstrates superior visual quality. Compared with state-of-the-art 3DGS-based low-light reconstruction methods LLGS and Luminance-GS, our approach exhibits superior capability in preserving high-frequency structural details. This is particularly evident in regions with extreme underexposure (e.g., the wardrobe area in the bedroom scene Figure 4), where RGB-only methods fail to recover texture due to extremely low RGB pixel values, whereas our fusion strategy leverages robust structural cues of the NIR modality to reconstruct meaningful scene content. In comparison with the 2D enhancement pipeline (DVN + Splatfacto), our method mitigates interview inconsistency artifacts. The per-frame enhancement in DVN introduces spatially varying luminance distortions, which propagates to 3D reconstruction as ghosting (e.g., evident in the living room scene Figure 5). When compared with the modified baseline Splatfacto_IHE, our framework achieves more effective cross-modal information integration, as evidenced by the high-fidelity reconstruction in the tractor scene (Figure 6). In particular, Splatfacto_IHE struggles to recover the Chinese characters engraved in the metal surface of the tractor, whereas our method reconstructs them clearly.

In summary, the above visual comparison confirms that our method achieves superior geometric accuracy, texture preservation, and artifact suppression in low-light 3D reconstruction compared with other methods. See SM Section S7 for reconstruction results of study, entrance hallway, HVAC room, and gate scenes.

3.2.2 | Comparison of Visible Edges and Average Gradient

The comparison results of visible edges and average gradient are summarized in Table 1. Our method achieves the highest newly visible edge retention ratio and average gradient ratio across all 7 scenes, indicating superior reconstruction and enhancement clarity in real-world low-light settings. Section S7 provides a comparison of the edge-recovery results.

3.2.3 | Comparison of High Quality Matching Number

The feature matching results are summarized in Table 2. NIRGB-GS achieves the highest number of high-quality matches across all 7 scenes, validating its potential to facilitate mobile robotic perception in low-light environments. Feature-matching visualizations are provided in Section S7.

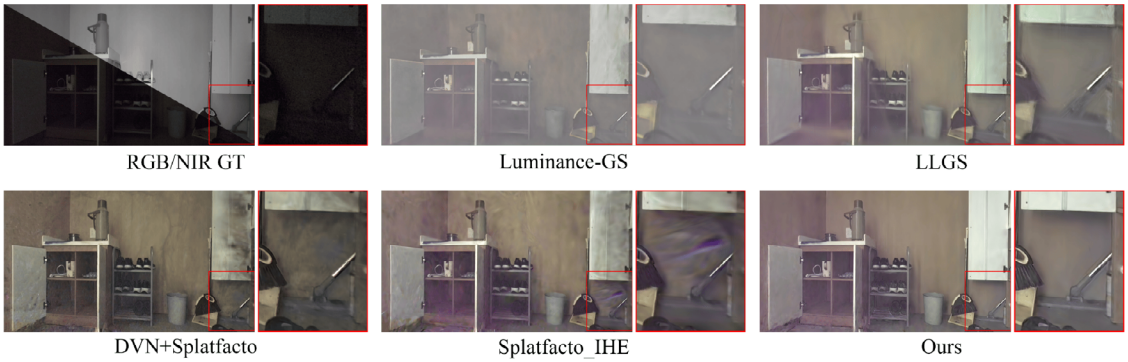


FIGURE 5 | Results of the living room scene. Since we leverage NIR opacity α_2 to dominate Gaussian pruning, floating objects and artifacts in the scene are significantly reduced compared to other methods.

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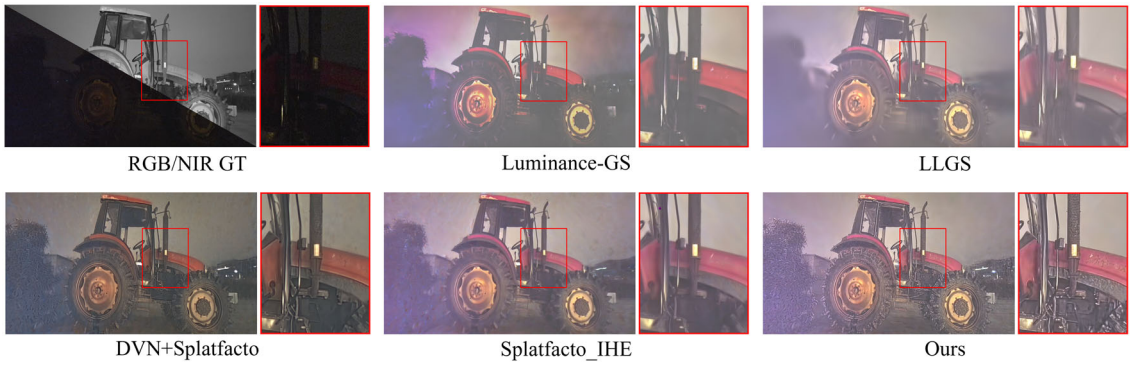


FIGURE 6 | Results of the tractor scene. Since our method effectively leverages the auxiliary NIR modality, it produces the clearest structural reconstructions. In the zoomed-in views, only our method recovers the engraved Chinese characters on the metal surface with clear outlines, achieving better visual quality than the other methods.

TABLE 1 | Comparison of newly visible edge ratio and average gradient ratio.

Metric	Method	Bedroom	Living room	Tractor	Study	Entrance hallway	HVAC room	Gate
Newly visible edge ratio ↑	Luminance-GS	2.655	4.070	8.195	15.418	7.919	1.396	0.414
	LLGS	2.489	3.690	7.624	14.000	10.599	2.247	−0.075
	DVN+Splatfacto	3.318	12.006	19.872	7.271	11.004	16.730	0.359
	Splatfacto_IHE	2.419	7.066	8.379	31.625	5.121	−0.083	0.264
	Ours	**5.181**	**20.614**	**60.133**	**57.366**	**84.053**	**32 .169**	**3.890**
Average gradient ratio ↑	Luminance-GS	1.426	2.431	1.734	2.127	1.084	0.883	1.799
	LLGS	1.302	2.126	1.502	1.867	0.976	1.496	1.316
	DVN+Splatfacto	1.363	2.017	1.677	1.689	0.773	0.999	1.237
	Splatfacto_IHE	1.259	2.284	1.343	2.001	0.854	0.883	1.487
	Ours	**1.656**	**3.204**	**3 000**	**2.466**	**1.671**	**1.649**	**1.833**

TABLE 2 | Comparison of feature matching.

Metric	Method	Bedroom	Living room	Tractor	Study	Entrance hallway	HVACroom	Gate
High quality matching number ↑	Luminance-GS	351	29	22	70	2	58	245
	LLGS	375	9	15	31	0	46	148
	DVN+Splatfacto	455	38	21	23	6	127	216
	Splatfacto_IHE	359	32	19	60	10	39	229
	Ours	**600**	**55**	**25**	**97**	**83**	**302**	**317**

3.3 | Hyperparameter Values and Ablation Study

In this section, we discuss the impact of hyperparameters on the reconstruction performance. We also conduct an ablation study to highlight the importance of the appearance model and the cross-modal loss $L_{\text{SSIM}-\text{NIR}}$.

3.3.1 | Hyperparameter Values

Without loss of generality, we first analyze the influence of λ_1. With λ_2 and λ_3 fixed at 0.2 and 1 respectively, the bedroom scene results taking $\lambda_1 = 1, 2, 3$ are shown in Figure 7.

When $\lambda_1 = 1$, the model's weight allocation to learning RGB color features is insufficient. NIR modality severely interferes with the RGB reconstruction, resulting in an overall whitish scene that clearly deviates from the color of normal visible scenes.

When $\lambda_1 = 3$, excessive emphasis on learning scene RGB color features hinders the effective integration of NIR information into the RGB reconstruction. This has a pronounced impact on regions that are completely dark in the RGB inputs, where pixel intensities are close to zero. For instance, the object structure near the horizontal bar beneath the wardrobe appears blurry.

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FIGURE 7 | Visualization of the effects employing different hyperparameter values λ_1.

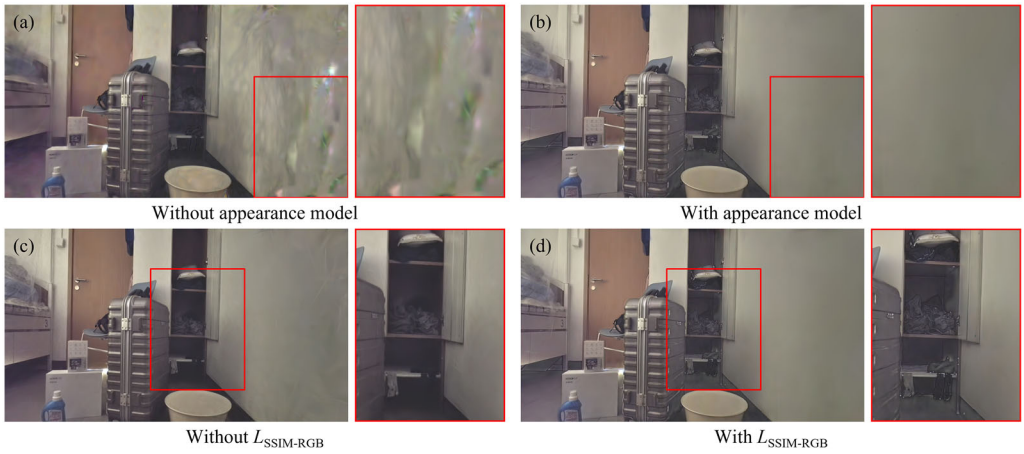


FIGURE 8 | Ablation studies of key components in our NIRGB-GS. (a,b) Ablation of the appearance model. (c,d) Ablation of the cross-modal SSIM loss.

When $\lambda_1 = 2$, a proper balance among the loss terms is achieved, leading to an effective fusion of color information in the RGB modality and structural information in the NIR modality. Thus, $\lambda_1 = 2$ is adopted in this article.

We further discuss the effects of λ_2 and λ_3 as well as their sensitivity to small variations in Section S4.

3.3.2 | Ablation Study

Without the appearance model, severe artifacts arise from NIR illumination variations and RGB white balance instability in low-light scenes, as shown by the distortions in Figure 8a. Incorporating the appearance modeling framework effectively mitigates these distortions, yielding a clear reconstruction as shown in Figure 8b.

Without the cross-modal SSIM loss between rendered normal-light images and NIR ground truth, the areas nearly completely dark in the RGB inputs cannot be properly reconstructed, such as the bottom of wardrobe in Figure 8c.

4 | Conclusion

In this article, we propose NIRGB-GS, a novel multimodal 3DGS-based method for scene reconstruction and enhancement under low-light conditions. Our approach addresses the fundamental limitations of existing RGB-only 3D reconstruction methods in dark scenes, such as failed geometry reconstruction and incomplete detail recovery, by exploiting the complementary strengths of NIR and RGB modalities. High-SNR NIR images are utilized to stabilize geometry estimation and mitigate the interference of RGB noise, while natural color ion is recovered through inverse histogram equalization. Comprehensive experiments across 7 real-world low-light environments demonstrate that NIRGB-GS significantly outperforms existing state-of-the-art methods in both visual quality and objective metrics. NIRGB-GS provides a methodology for effectively fusing multimodal information for high-performance 3D reconstruction of low-light scenarios, with promising applications in robotics and augmented reality systems operating in poorly illuminated environments.

Author Contributions

Chengyun Yang: methodology, software, validation, formal analysis, investigation, data curation, writing – original draft, visualization. **Yi Zhang**: methodology, software, validation, formal analysis, investigation, data curation. **Yu Lei**: resources, writing – review and editing, supervision. **Qiaofeng Li**: conceptualization, resources, writing – review and editing, supervision, project administration, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Figure S1**: RGB and NIR COLMAP results comparison. (a) COLMAP fails on the RGB images of HVAC room scene, due to noise and insufficient brightness. (b) COLMAP successfully recovers camera poses and initial sparse points on corresponding NIR images of the same scene. At the bottom are example input views for RGB and NIR, respectively. **Supporting Figure S2**: RGB dataset color inconsistency. Two low-light RGB frames (a) and (b) from the entrance hallway scene exhibit noticeable color differences: frame (a) has a distinctly stronger blue tone than frame (b). The corresponding RGB channel histograms confirm this observation, while the red and green channel means differ only slightly, the blue channel mean of (a) (39.49) is substantially higher than that of (b) (27.34). **Supporting Figure S3**: The appearance model in our NIRGB-GS framework for obtaining color features of NIR and RGB modalities. **Supporting Figure S4**: Bedroom dataset pixel distribution. The above images show the pixel distribution of our bedroom dataset (collection of 41 images). The threshold covering 99 percent of pixels is from 15 to 92. Based on this range, the upper threshold k_{max} is determined to be 92. **Supporting Figure S5**: Normal-light reference images for color recovery assessment. For the bedroom and living room scenes, we compare our enhanced reconstruction results against images of the same scenes captured under sufficient lighting. The close color correspondence indicates effective color recovery. **Supporting Figure S6**: Effects of λ_1, λ_2, and λ_3 on the bedroom scene. First row: $\lambda_1 \in \{1, 2, 3\}$ with $\lambda_2 = 0.2$ and $\lambda_3 = 1$. Second row: $\lambda_2 \in \{0, 0.2, 0.5\}$ with $\lambda_1 = 2$ and $\lambda_3 = 1$. Third row: $\lambda_3 \in \{0.5, 1, 2\}$ with $\lambda_1 = 2$ and $\lambda_2 = 0.2$. **Supporting Figure S7**: Sensitivity analysis under small weight variations on the bedroom scene. From left to right, we vary λ_1 in {1.9, 2, 2.1}, λ_2 in {0.19, 0.2, 0.21}, and λ_3 in {0.9, 1, 1.1} (other weights fixed to the adopted defaults). The results are visually consistent, indicating robustness of the proposed loss weighting. **Supporting Figure S8**: Sensitivity analysis under small weight variations on the study scene, using the same parameter variations as Figure S7. These results further confirm the robustness of NIRGB-GS to small loss weight variations. **Supporting Figure S9**: Shared v.s. separate position and covariance across modalities. We show results on the bedroom and study scenes. Sharing position and covariance (ours) reduces the impact of noisy low-light RGB inputs on geometry, leading to fewer floating artifacts (bedroom) and clearer reconstructions with fewer artifacts (study). This shared-geometry design reduces parameter redundancy and helps prevent noisy low-light RGB observations from per- turbing the underlying geometry. **Supporting Figure S10**: Discussion of the percentile used for IHE on the living room scene. We compare using the 40th, 99th, and 99.99th percentiles to compute k_{max}. Using the 40th percentile leads to overexposed reconstructions and loss of texture datails. while using the 99.99th percentile makes the reconstruction excessively dark. The adopted 99th percentile achieves a balanced brightness while preserving details. **Supporting Figure S11**: Results of the study scene. The top left image is the RGB/NIR GT for the reconstructed scene. Each image incorporates a magnified view of the region highlighted by the red box. In this case, our method is the only approach that clearly reconstructs power cords in extremely dark areas, due to its effective integration of structural information from NIR images. **Supporting Figure S12**: Results of the entrance hallway scene. Our method can clearly reconstruct the text and patterns on the fire extinguisher box (enlarged in red box), while other methods produce blurred structures or suffer from fioating artifacts in this region due to inherent noise in low-light RGB images. **Supporting Figure S13**: Results of the HVAC room scene. In this case, our method achieves the clearest reconstruction in the label area, whereas other methods exhibit varying degrees of artifact obstruction. **Supporting Figure S14**: Results of the outdoor gate scene. In particular, this scene is partially illuminated by the street lamps behind the bars, leading to diverse and spatially varying low-light conditions. Even under such varied illumination, our method remains robust and produces stable, high-quality reconstructions by effectively leveraging the auxiliary NIR modality. As a result, it achieves the clearest structural reconstruction. In the enlarged views, our method recovers tree branches and leaves with sharp outlines, achieving better visual quality than the other methods. **Supporting Figure S15**: Edge detection results for the living room scene. These images show the edge detection results from novel view synthesis images (the same perspective as Figure 5 in the main paper). As our method effectively suppresses artifacts and clearly reconstructs scene structures, it extracts the most valid edge information. **Supporting Figure S16**: Evaluation of Feature Matching on our HVAC room scene. Our method achieves a substantially higher high-quality matching number compared to alternative approaches.

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